

Assignment 2

ARMA Processes and Seasonal Processes

Group 49

Jiayue Che, s252639

Qiaowen You, s252709

Anqi Yu, s254721

Contents

1. Stability 3

1.1 Stationarity Check 3

1.2 Invertibility 3

1.3 Autocorrelation ρ(k) for AR(2) 3

1.4 Plot ρ(k) up to lag 30 4

2. Simulating Seasonal Processes 4

2.1 A (1,0,0) $\times(0,0,0)_{12}$ model with the parameter $\phi_1 = 0.6$. 4

2.2. A (0,0,0) $\times(1,0,0)_{12}$ model with the parameter $\Phi_1 = -0.9$ 5

2.3. A (1,0,0) $\times(0,0,1)_{12}$ model with the parameters $\phi_1 = 0.9$ and $\Theta_1 = -0.7$ 6

2.4. A (1,0,0) $\times(1,0,0)_{12}$ model with the parameters $\phi_1 = -0.6$ and $\Phi_1 = -0.8$ 7

2.5. A (0,0,1) $\times(0,0,1)_{12}$ model with the parameters $\theta 1 = 0.4$ and $\Theta_1 = -0.8$ 8

2.6. A (0,0,1) $\times(1,0,0)_{12}$ model with the parameters $\theta_1 = -0.4$ and $\Phi_1 = 0.7$ 9

2.7 Summary 10

3 .Identifying arma models 12

3.1 Process 1 12

3.2 Process 2 12

3.3 Process 3 12

1. Stability

1.1 Stationarity Check

The AR(2) process is: $X_t + \phi_1 X_{t-1} + \phi_2 X_{t-2} = \epsilon_t$

With $\phi_1 = -0.7$, $\phi_2 = -0.2$, rewrite as: $X_t = 0.7X_{t-1} + 0.2X_{t-2} + \epsilon_t$

The characteristic equation is: $1 - 0.7z - 0.2z^2 = 0 \Rightarrow 0.2z^2 + 0.7z - 1 = 0$

Solve for roots:

$$Z = \frac{-0.7 \pm \sqrt{0.7^2 - 4 \times 0.2 \times 1}}{2 \times 0.2} = \frac{-0.7 \pm \sqrt{1.29}}{0.4}$$

$$z_1 = \frac{-0.7 + 1.136}{0.4} \approx 1.09,\ z_2 = \frac{-0.7 - 1.136}{0.4} \approx -4.59$$

For stationarity, all roots must lie outside the unit circle ($|z|>1$):

$|z_1| \approx 1.09 > 1$, $|z_2| \approx 4.59 > 1$.

So, the process is stationary.

1.2 Invertibility

For an AR(p) process, the error term ϵ_t can be expressed in terms of X_t and its past values.

Therefore, the process is invertible.

1.3 Autocorrelation ρ(k) for AR(2)

The model can be rewritten as $X_t = 0.7X_{t-1} + 0.2X_{t-2} + \epsilon_t$

The Yule-Walker equations give:

$$\rho(1) = \frac{\phi_1}{1 - \phi_2}$$

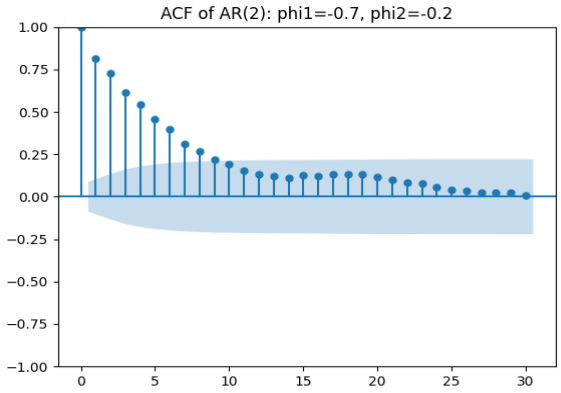
$$\rho(2) = \phi_1 \rho(1) + \phi_2$$

$$\rho(k) = \phi_1 \rho(k-1) + \phi_2 \rho(k-2), k \geq 3$$

For $k \geq 3$, ρ(k) follows the recurrence:

$$\rho(k) = 0.7\rho(k-1) + 0.2\rho(k-2)$$

1.4 Plot ρ(k) up to lag 30



Using the recurrence relation, ρ(k) can be computed for k=0, …, 30.

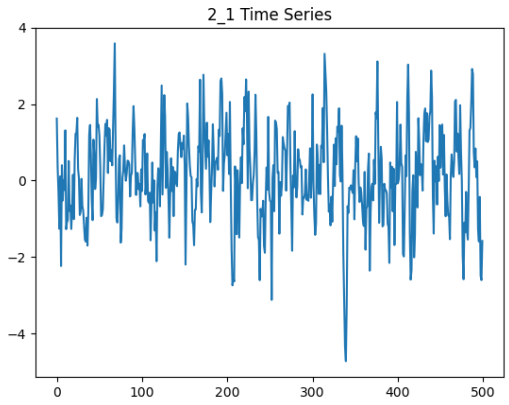
The ACF decreases gradually toward zero, showing an exponential decay pattern.

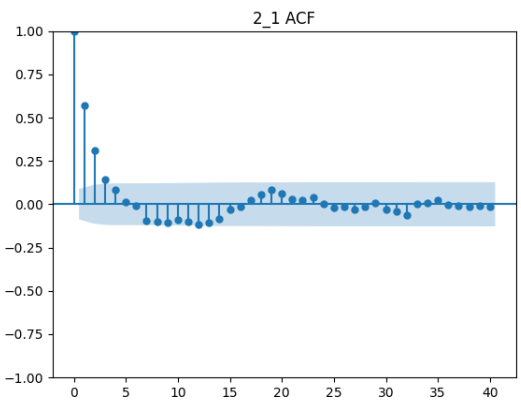
This behavior is characteristic of a stationary AR(2) process, where correlations persist but diminish over time.

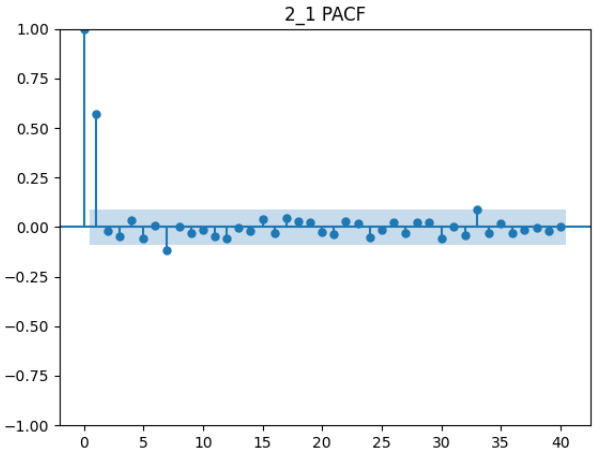
2. Simulating Seasonal Processes

2.1 A (1,0,0) ×(0,0,0)$_{12}$ model with the parameter $\phi_1 = 0.6$.

Model: $Y_t = 0.6Y_{t-1} + \epsilon_t$ (non-seasonal AR(1))







Time Series: The series exhibits a smooth, positively correlated pattern typical of an AR(1) process. Values tend to persist, with no evident seasonal fluctuations.

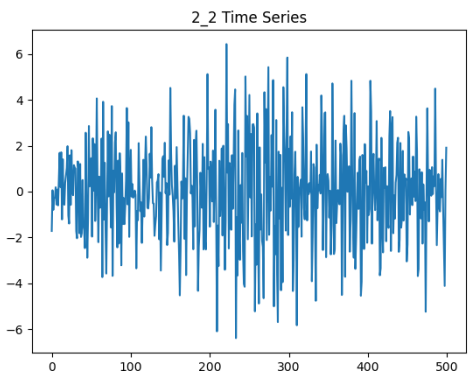
ACF: Shows exponential decay from lag 1 onward. All autocorrelations remain positive and decay gradually, characteristic of a stationary AR(1) process with $\phi 1 = 0.6 \phi 1 = 0.6$.

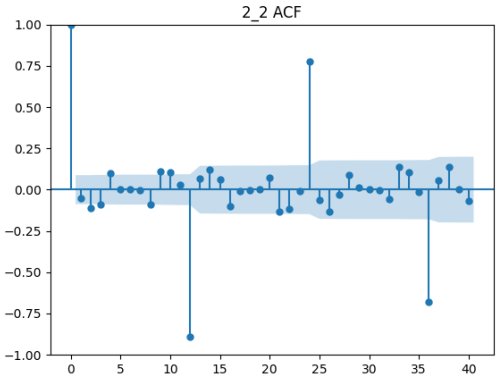
PACF: Cuts off sharply after lag 1. Only the first partial autocorrelation is significant, confirming the AR(1) structure.

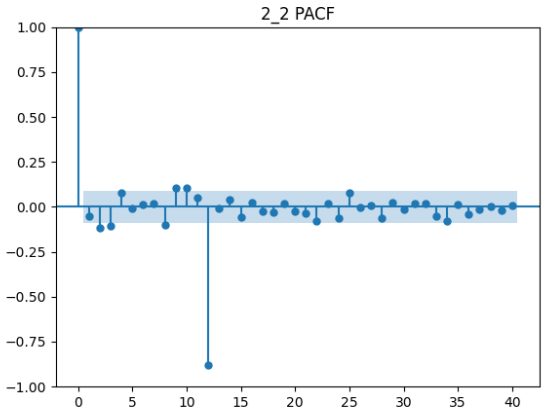
Conclusion: This is a pure non-seasonal AR(1) process with positive autocorrelation, leading to a smooth, persistent time series.

2.2. A $(0,0,0) \times (1,0,0)_{12}$ model with the parameter $\Phi_1 = -0.9$

Model: $Y_t = -0.9 Y_{t-12} + \epsilon_t$ (seasonal AR(1) with s=12)







Time Series: Displays clear 12-period cycles with alternating peaks and troughs. The negative seasonal AR parameter (−0.9−0.9) creates a strong "see-saw" pattern where high values tend to be followed by low values 12 periods later.

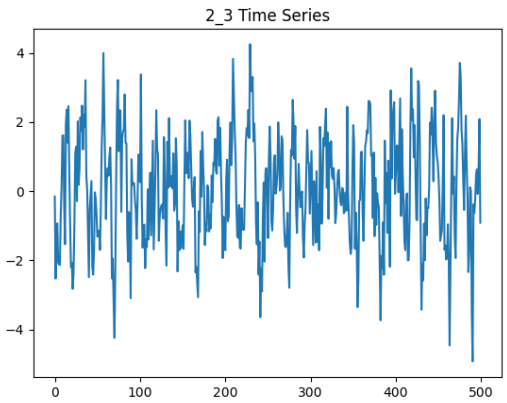
ACF: Exhibits significant negative spikes at lags 12, 24, 36, etc. The pattern decays slowly, with alternating signs at seasonal lags. Non-seasonal lags show minimal correlation.

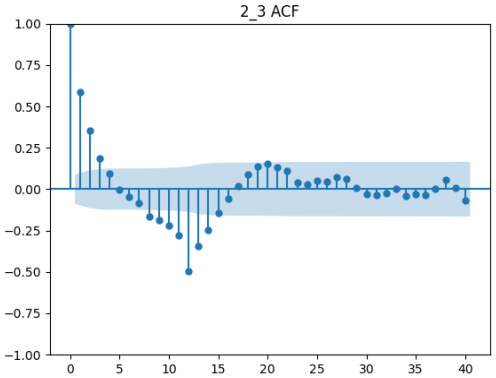
PACF: Shows a single significant negative spike at lag 12, with no other significant partial autocorrelations. This confirms the pure seasonal AR(1) structure.

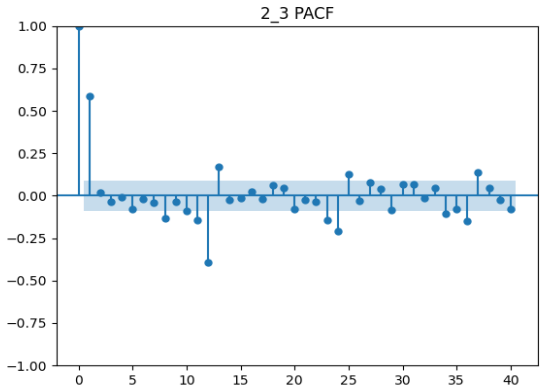
Conclusion: This is a pure seasonal AR(1) process with strong negative seasonal dependence, creating a regular oscillatory pattern at the seasonal frequency.

2.3. A $(1,0,0) \times (0,0,1)_{12}$ model with the parameters $\phi_1 = 0.9$ and $\Theta_1 = -0.7$

Model: $Y_t = 0.9Y_{t-1} + \epsilon_t + 0.7\epsilon_{t-12}$ (AR(1) + seasonal MA(1))







Time Series: Shows a combination of smooth, persistent movements (due to $\phi 1 = 0.9 \phi 1 = 0.9$) with 12-period adjustments (due to $\phi 1 = -0.7\ \phi 1 = -0.7$). The series is relatively smooth overall.

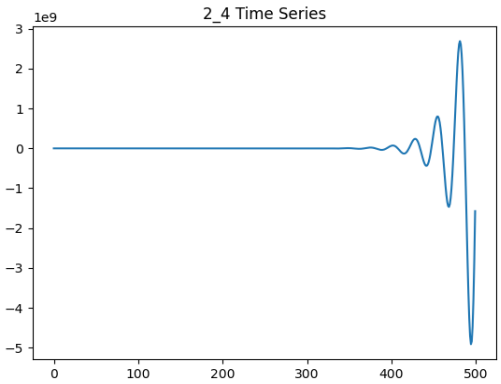
ACF: Displays exponential decay from lag 1 onward (AR component), with an additional significant negative spike at lag 12 (seasonal MA component). The decay continues beyond lag 12 but with a distinctive pattern at seasonal multiples.

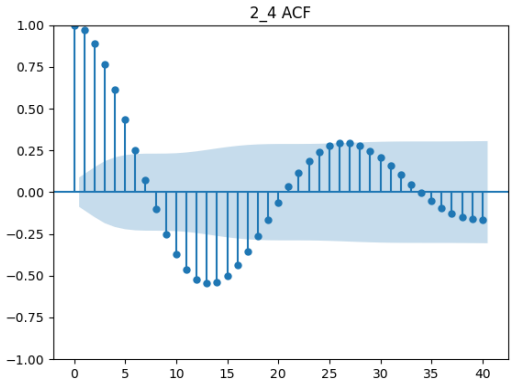
PACF: Shows a significant spike at lag 1 (AR component) and then decays. There may be additional patterns at lag 12, but the AR(1) dominates the partial autocorrelation structure.

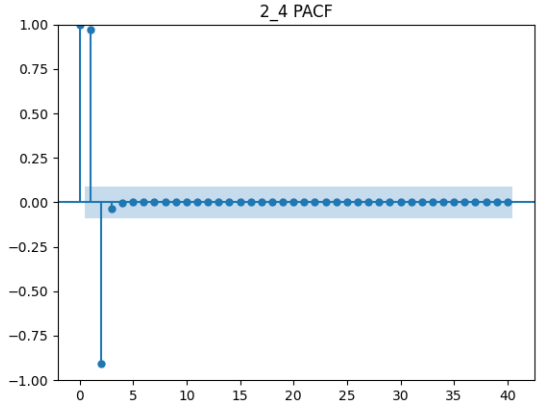
Conclusion: This is a mixed model where the AR(1) component creates short-term persistence, while the seasonal MA(1) component introduces a 12-period adjustment. The negative $\Theta 1 \Theta 1$ induces a negative correlation at lag 12.

2.4. A $(1,0,0) \times (1,0,0)_{12}$ model with the parameters $\phi_1 = -0.6$ and $\Phi_1 = -0.8$

Model: $Y_t = -0.6 Y_{t-1} - 0.8 Y_{t-12} + 0.48 Y_{t-13} + \epsilon t$ (multiplicative AR(1)×AR(1))







Time Series: Exhibits both short-term oscillatory behavior (due to negative $\phi 1\phi 1$) and longer 12-period cycles (due to negative $\Phi 1\Phi 1$). The series alternates frequently at both scales.

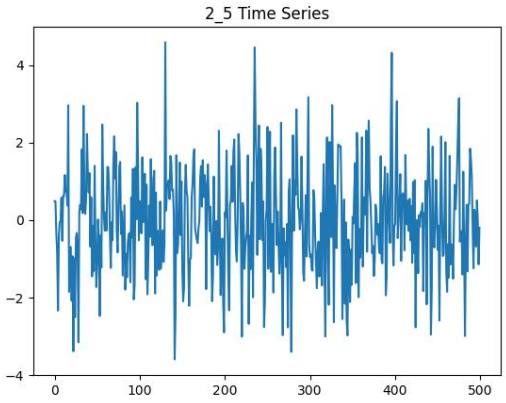
ACF: Shows a damped sinusoidal pattern at low lags (from $\phi 1 = -0.6\phi 1 = -0.6$) and additional significant spikes at lags 12, 24, etc. (from $\Phi 1 = -0.8\Phi 1 = -0.8$). The pattern is a superposition of both AR components.

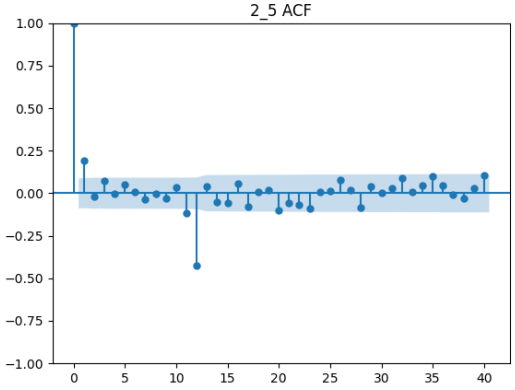
PACF: Exhibits significant spikes at lag 1 (from $\phi 1\phi 1$) and lag 12 (from $\Phi 1\Phi 1$), with possible interaction at lag 13. The partial autocorrelation structure reflects both AR components clearly.

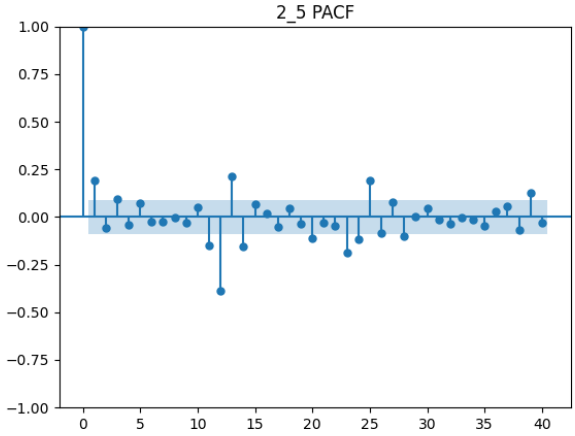
Conclusion: This is a multiplicative seasonal AR model with both regular and seasonal AR components. The negative coefficients create oscillatory patterns at multiple frequencies.

2.5. A $(0,0,1) \times (0,0,1)_{12}$ model with the parameters $\theta 1 = 0.4$ and $\Theta_1 = -0.8$

Model: $Yt = \epsilon t - 0.4\epsilon_{t-1} + 0.8\epsilon_{t-12} - 0.32\epsilon_{t-13}$ (multiplicative MA(1)×MA(1))







Time Series: Appears more erratic than AR-based models, with both short-term and seasonal patterns visible. The series shows some 12-period structure with shorter fluctuations.

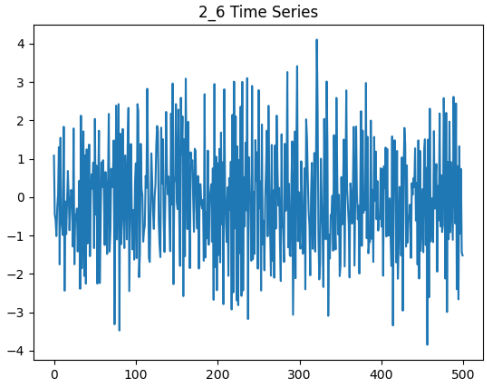
ACF: Cuts off after lag 1 (due to $\theta 1 = 0.4\theta 1 = 0.4$) and after lag 12 (due to $\Theta 1 = -0.8\Theta 1 = -0.8$). There is a significant positive spike at lag 1 and a significant negative spike at lag 12. The ACF is zero beyond these lags.

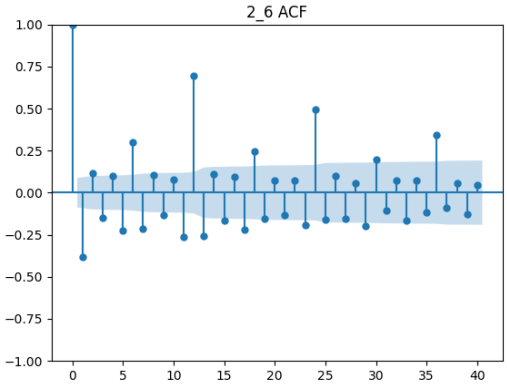
PACF: Exhibits exponential decay rather than a sharp cutoff, characteristic of an MA process. The decay includes both short-term and seasonal patterns.

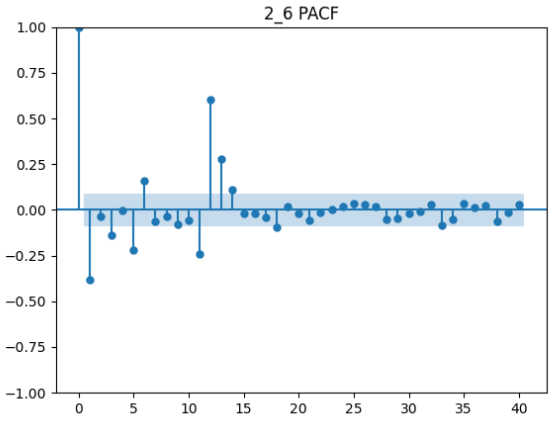
Conclusion: This is a pure seasonal MA model with both regular and seasonal MA components. The ACF provides direct identification of the MA orders (cutoffs at lags 1 and 12).

2.6. A $(0,0,1) \times (1,0,0)_{12}$ model with the parameters $\theta_1 = -0.4$ and $\Phi_1 = 0.7$

Model: $Yt = 0.7Y_{t-12} + \epsilon t + 0.4\epsilon_{t-1} - 0.28\epsilon_{t-13}$ (MA(1) + seasonal AR(1))







Time Series: Displays a combination of short-term negative autocorrelation (due to $\theta 1 = -0.4\theta 1 = -0.4$) and positive 12-period seasonal persistence (due to $\Phi 1 = 0.7\Phi 1 = 0.7$). The series shows some oscillatory behavior at short lags and smoother seasonal cycles.

ACF: Shows a significant negative spike at lag 1 (from $\theta 1 = -0.4\theta 1 = -0.4$) and then decays. At seasonal lags, positive spikes appear at 12, 24, etc., decaying gradually due to the seasonal AR component.

PACF: Exhibits a sharp cutoff after lag 1 (from the MA component) and then additional patterns at seasonal lags consistent with the seasonal AR(1) structure.

Conclusion: This mixed model combines an MA(1) short-term component with a seasonal AR(1) component. The ACF reflects both: a cutoff at lag 1 followed by seasonal persistence.

2.7 Summary

Key Observations:

Model	Structure	ACF Characteristics	PACF Characteristics
2.1	AR(1)	Exponential decay	Cutoff at lag 1
2.2	Seasonal AR(1)	Spikes at 12, 24, 36...	Cutoff at lag 12
2.3	AR(1) × Seasonal MA(1)	Exponential decay + spike at 12	Spike at 1 + decay
2.4	AR(1) × Seasonal AR(1)	Damped oscillations at low lags + spikes at seasonal lags	Spikes at 1 and 12
2.5	MA(1) × Seasonal MA(1)	Cutoff at 1 and 12	Exponential decay at low and seasonal lags
2.6	MA(1) × Seasonal AR(1)	Cutoff at 1 + seasonal spikes	Cutoff at 1 + seasonal patterns

General Behavior of ACF for Seasonal Processes:

1. Pure Seasonal AR Processes: The ACF shows significant spikes at seasonal lags (s, 2s, 3s, ...) that decay exponentially, while the PACF cuts off after the seasonal lag s.
2. Pure Seasonal MA Processes: The ACF cuts off after the seasonal lag s, while the PACF shows exponential decay at seasonal lags.
3. Multiplicative Models: The ACF combines patterns from both components:
    a. Short-term patterns (from non-seasonal component) appear at lags 1, 2, ...
    b. Seasonal patterns appear at lags s, 2s, 3s, ...
    c. Interaction effects may appear at lags s+1, s+2, etc.
4. Sign of Coefficients:
    a. Positive coefficients lead to positive autocorrelations at the corresponding lags
    b. Negative coefficients create alternating patterns or negative spikes
5. Model Identification: By examining both ACF and PACF together, one can identify:
    a. Whether non-seasonal components are AR (PACF cutoff) or MA (ACF cutoff)
    b. Whether seasonal components are AR (PACF spikes at seasonal lags) or MA (ACF spikes at seasonal lags)
    c. The order of both non-seasonal and seasonal components
6. Practical Implication: For real data analysis, seasonal patterns in ACF/PACF that persist

at multiples of the seasonal period (e.g., 12, 24, 36 for monthly data) indicate the need for seasonal ARIMA modeling.

3 .Identifying arma models

To identify the appropriate model, we analyze the behavior of the autocorrelation function (ACF) and the partial autocorrelation function (PACF):

For an AR(p) process: ACF → gradually decays. PACF → cuts off after lag p

For an MA(q) process: ACF → cuts off after lag q. PACF → gradually decays

For an ARMA(p, q) process: Both ACF and PACF → gradually decay

3.1 Process 1

The ACF and PACF show no significant structure, with most values within the confidence bounds. This indicates that the series is white noise.

Therefore, the process is ARMA(0,0).

3.2 Process 2

The ACF decays gradually, while the PACF cuts off after lag 1. This pattern is characteristic of an AR process.

Therefore, the process is likely AR(1).

3.3 Process 3

Both the ACF and PACF decay gradually without a clear cutoff. This indicates that the process cannot be modeled as a pure AR or MA model, since neither function shows a sharp cutoff.

Therefore, the process is an ARMA(p,q).